RETAIL CUSTOMER BEHAVIOR & PURCHASE PATTERN ANALYTICS REPORT

An End-to-End Case Study on Data Preprocessing, Relational Database Migration, SQL Querying, and BI Dashboarding.

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1. The Business Problem & Objectives

Context:

In modern retail and e-commerce markets, understanding consumer purchase behavior is essential for sustaining growth. Without structured data analytics, businesses make operational decisions based on intuition rather than evidence, leading to margin-eroding promotions, misallocated inventory, and friction during the checkout process. This project addresses the lack of visual analytics and data structure.

Core Business Problems Addressed:

- **Unoptimized Promotions:** Are discount codes driving larger order sizes, or are they simply diluting profits?
- **Seasonal Demand Misalignment:** How do product categories perform across different seasons, and how can we prevent storage overheads?
- **Payment Option Friction:** What checkout payment methods do different demographic cohorts prefer?
- **Customer Segmentation:** Do subscribed members exhibit higher lifetime value (LTV) than non-subscribers?

Project Deliverables:

1. Develop a Python preprocessing pipeline using Pandas to clean transactional data.
2. Migrate processed records to a local PostgreSQL relational database server.
3. Execute SQL analytical scripts to segment customers and answer business questions.
4. Design publication-grade visuals (Seaborn) and an interactive Power BI dashboard.

2. Methodology & Implementation ("What We Have Done")

To solve the business objectives, an end-to-end data engineering and analytics pipeline was constructed consisting of four distinct stages: ingestion/cleaning, database storage, querying, and interactive visualization.

Stage 1: Python Data Cleaning & Preprocessing

Using Python and Pandas, the raw transaction dataset (3,900 rows) was loaded and formatted. Column headers were converted to snake_case. Missing data in the 'Review Rating' field was handled using category-median imputation to preserve record volume. Feature engineering was performed to create 'age_group' buckets and calculate purchase frequencies in numerical days.

Stage 2: Relational Database Migration

To enable secure, structured querying, the cleaned pandas DataFrame was migrated directly into a local PostgreSQL database. The connection was programmatically established using SQLAlchemy and psycopg2. This mimics a professional corporate environment where data is pulled directly from the database rather than flat files.

Stage 3: SQL Analytics Execution

Using a database client (DBeaver), optimized SQL scripts were written to execute analytics directly against the database table. Queries were written to calculate revenue ratios, identify loyal segments, and output metrics for reporting.

Stage 4: Visualization & BI Dashboarding

A dual-visualization layout was developed. Matplotlib and Seaborn were used inside the Jupyter Notebook for immediate visual analysis. Subsequently, a Power BI dashboard was connected via local network to the PostgreSQL database, enabling executives to filter and slice transactional metrics interactively.

3. Detailed Findings & Strategic Recommendations

A. Discount Program Impact Analysis

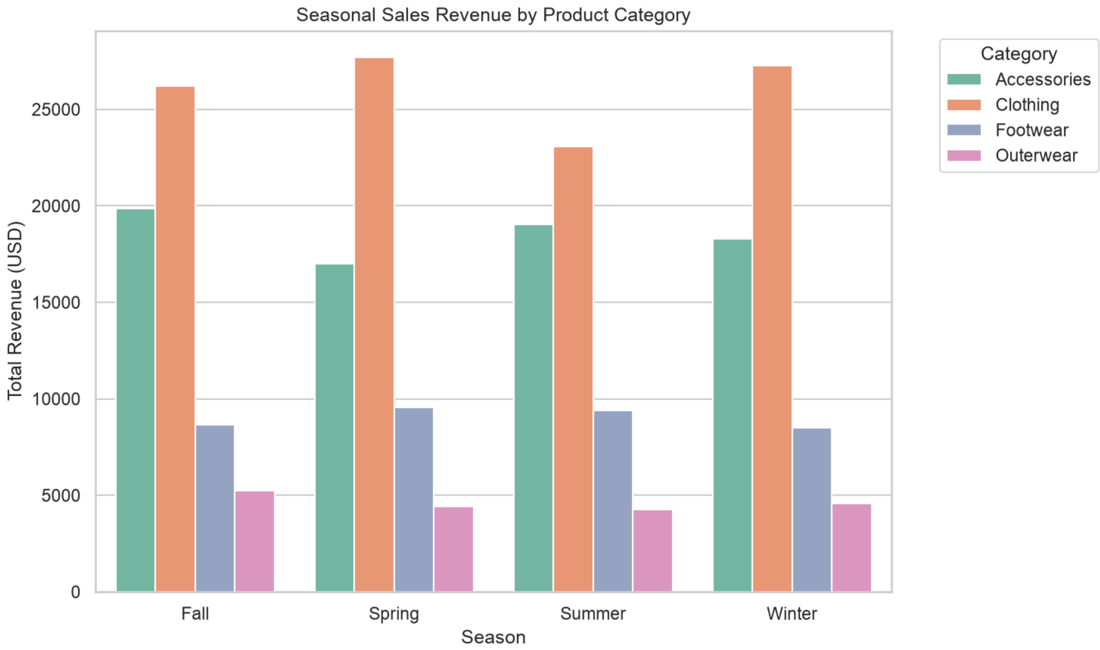
Business Findings: The analysis reveals that the average transaction spend for customers utilizing discounts was $59.28. In contrast, customers who did not use discounts spent $60.13. Offering discounts failed to drive higher cart volumes or overall transaction sizes.

Strategic Suggestions: The discount program is currently diluting net profit margins. It is recommended to shift from flat discount rates to threshold-based incentives (e.g., 'Receive 15% off when you spend $75 or more'). This structures the discount to push the customer to add more items to their cart, raising the Average Order Value (AOV).

B. Seasonal Demand Analysis (Inventory Optimization)

Business Findings: Different product categories exhibit heavy seasonal revenue shifts. 'Clothing' sales are extremely high in Spring ($27.7k) and Winter ($27.3k), but drop significantly in Summer ($23k). Conversely, 'Accessories' generate peak sales in Summer and Fall.

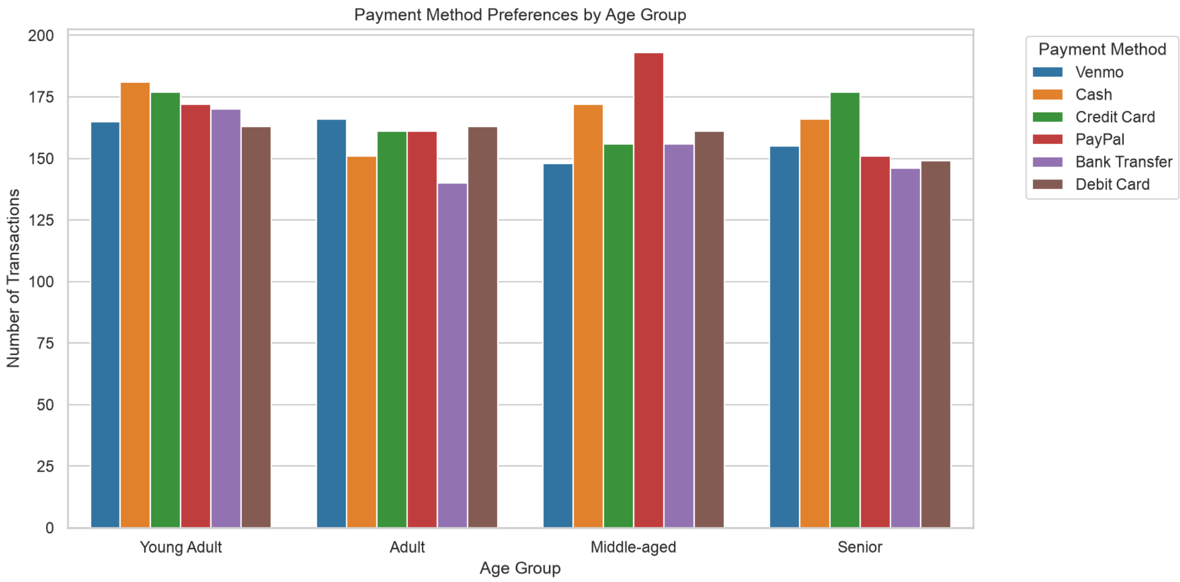
Strategic Suggestions: To reduce holding and warehousing costs, implement dynamic space reallocation. Scale down clothing inventory in the summer and allocate that shelf space to accessories, while prioritizing clothing stock replenishment ahead of the winter season.



C. Demographic Checkout Preferences

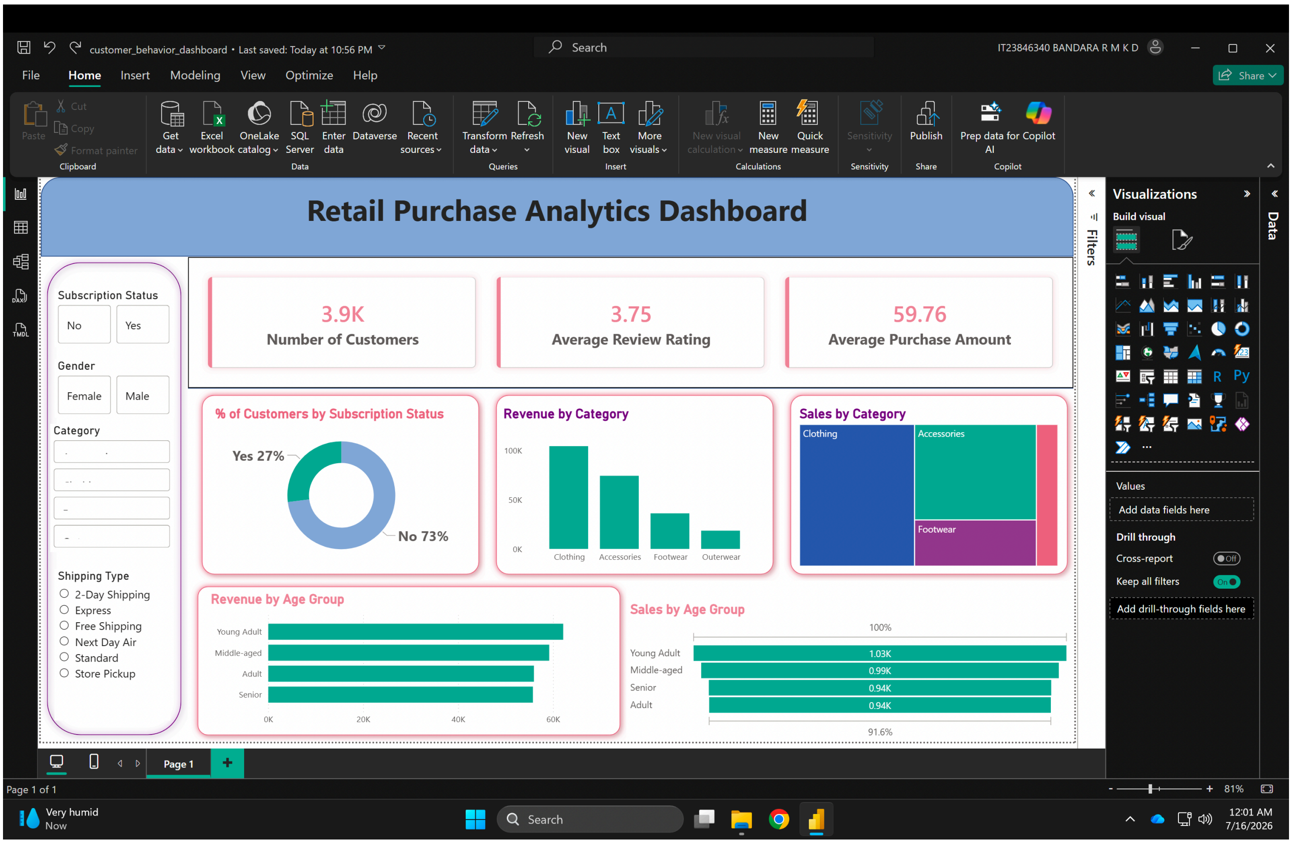
Business Findings: Slicing payment methods across age cohorts shows that middle-aged customers exhibit a heavy preference for PayPal, while seniors and young adults rely mostly on Credit Cards and Cash.

Strategic Suggestions: Surfacing unnecessary payment options creates cognitive friction at checkout. By implementing payment options that highlight PayPal for middle-aged users and card processors for other groups, we can create a frictionless checkout flow, directly reducing cart abandonment rates.



4. Interactive Executive Reporting Dashboard

To enable real-time visual decision-making, an interactive dashboard was built in Power BI Desktop. Instead of loading static flat files, the dashboard is linked directly to the local PostgreSQL database server, providing a robust enterprise reporting environment. The layout features real-time KPI metrics, category treemaps, age-group funnels, and slicers to filter sales metrics by customer demographics and shipping types.



5. Conclusion & Business Value Delivery

By constructing an end-to-end data pipeline and interactive BI dashboard, this project successfully bridges the gap between database operations and executive business decisions.

Key Business Value Delivered:
1. Identified that current discount schemes erode profit margins, recommending a spend-threshold model to increase Average Order Value.
2. Provided an actionable framework for inventory optimization based on seasonal category shifts, saving on warehousing costs.
3. Outlined payment-preference demographics to guide UX adjustments that reduce checkout cart abandonment.
4. Demonstrated professional data engineering principles by connecting Python pipelines to a relational PostgreSQL database and Power BI.

This project stands as a ready-to-deploy template for small-to-medium retail operations aiming to establish data-driven decision workflows.

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GitHub Code: https://github.com/your-username/retail-shopping-behavior-analysis
Powered by: Python, PostgreSQL, and Power BI